Assessing the expressive power of Graph Neural Networks

Ameni Jomaa
Student Researcher
ameni.jomaa@ensea.fr

Dr. Guillaume Renton
Supervisor
guillaume.renton@ensea.fr

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Abstract

As artificial intelligence and data-driven research evolve, an increasing amount of real-world information is being modeled as graph-structured data, spanning a wide variety of domains and topologies. This evolution has motivated significant progress in the development of Graph Neural Networks (GNNs), with diverse architectures proposed across both spatial and spectral paradigms. Despite these advancements, most existing models are built upon the homophily assumption, which inherently constrains their spectral behavior to low-pass or high-pass filtering. This design bias limits their expressive capacity when applied to graphs with low homophily, such as heterophilic structures. This study investigates the expressive capacity of GNNs through a spectral lens, with a focus on recent graph transformer architectures. We aim to assess whether this architecture can be adapted to exhibit band-pass filtering behavior which means is capable of capturing a broader range of frequency information and identify which components, such as positional encoding, LayerNormalization or the attention mechanism itself, most contribute to this capability. Through extensive ablation studies on both homophilic and heterophilic datasets, our results show that transformer-based models enhanced with structural encodings and normalization outperform traditional GNNs in heterophilic settings while remaining competitive in homophilic one.

1 Introduction

Graph Neural Networks (GNNs) have become a cornerstone of modern machine learning on structured data, enabling deep representation learning on graphs across diverse applications such as social network analysis, recommendation systems, molecular modeling, and traffic forecasting. A variety of architectures incorporating both spatial and spectral domains have been proposed [1, 2, 5]), each with different mechanisms for aggregating and propagating information over graph structures. Despite the variety of architectures, a key limitation persists which is the assumption of homophily, where connected nodes tend to share similar labels. As a result, they often behave as low-pass filters in the spectral domain effectively smoothing node features and suppressing high-frequency components. While this behavior is effective in homophilic settings, it severely limits the expressiveness of GNNs on heterophilic graphs, where adjacent nodes may belong to different classes. This has raised important questions about the generalization capacity of current GNN architectures and their ability to model structurally diverse graphs. In this research, we aim to analyze and enhance the expressive power of GNNs from a spectral filtering perspective, focusing on how different filtering behaviors influence model performance across graphs with varying levels of homophily and heterophily. Motivated by the observation that most GNNs operate as low-pass filters, we investigate the potential of graph transformer-based architectures to overcome these limitations by enabling band-pass filtering, which captures both local and global structural variations. Specifically, we explore two main research questions:

- What filtering techniques are best suited for graphs exhibiting different levels of homophily and heterophily?
- How can transformer-based architectures be effectively adapted to perform band-pass filtering, dynamically adjusting their focus to better capture the underlying graph structure?

To address these questions, we conduct a series of empirical evaluations and ablation study on representative homophilic and heterophilic datasets. Our goal is to identify the architectural components such as positional encoding, attention mechanism , and normalization, that most contribute to improved generalization across diverse graph types.

2 State of the Art

2.1 Spectral Foundation of GNNs

The spectral interpretation of Graph Neural Networks (GNNs) originates from principles in graph signal processing, where node features are viewed as signals residing on the graph's topology. This framework relies on the spectral decomposition of the graph Laplacian, defined as $L = D - A$, where D denotes the degree matrix and A represents the adjacency matrix. Through eigen-decomposition, the Laplacian can be expressed as:

$$L = U\Lambda U^\top \tag{1}$$

In this formulation, the columns of U correspond to the orthonormal eigenvectors of L, and Λ is a diagonal matrix containing the associated eigenvalues, which represent graph frequencies. Spectral convolution operations on graphs are then defined in the frequency domain by applying filter functions directly to these eigenvalues:

$$g_\theta \star x = U g_\theta(\Lambda) U^\top x \tag{2}$$

To make this operation more computationally tractable, Chebyshev Networks (ChebNet) [10] approximate the filter $g_\theta(\Lambda)$ using Chebyshev polynomials of the scaled Laplacian:

$$g_\theta(L) \approx \sum_{k=0}^{K} \theta_k T_k(\tilde{L}) \tag{3}$$

where $\tilde{L} = \frac{2L}{\lambda_{\max}} - I$ rescales the Laplacian for stability and efficiency. This polynomial approximation enables localized spectral filtering without explicitly computing the eigendecomposition, which is central to our subsequent investigation into filter design strategies that adapt to different levels of homophily and heterophily in graph data.

2.2 Spatial Foundations of GNNs

The spatial formulation of Graph Neural Networks, particularly those based on the Message Passing Neural Network (MPNN) framework, operates by iteratively updating node representations through the aggregation of local neighborhood information. At each layer, a node receives messages from its neighbors via a message function M, followed by an update function U that combines the incoming information with the node's current state:

$$m_v^{(l)} = \bigoplus_{u \in \mathcal{N}(v)} M(h_v^{(l-1)}, h_u^{(l-1)}, e_{uv}) \tag{4}$$

$$h_v^{(l)} = U(h_v^{(l-1)}, m_v^{(l)}) \tag{5}$$

Depending on the aggregate and update function of the MPNN we can have different types of Graph Neural Networks. The aggregation operator $\bigoplus$; which can be sum, mean, or maximum, plays a crucial role in determining the model's ability to capture structural information. For instance, the Graph Isomorphism Network (GIN) [3] demonstrates that by utilizing sum aggregation with injective functions achieves the expressive power of the Weisfeiler-Lehman (WL) test, making it one of the most powerful spatial GNNs in terms of theoretical discrimination.

Building on this foundation, attention-based mechanisms have been introduced to allow for adaptive, learnable neighborhood weighting. Graph Attention Networks (GATs) [4] compute attention coefficients that determine the importance of neighboring nodes:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^\top [Wh_i \,||\, Wh_j]))}{\sum_{k\in\mathcal{N}(i)} \exp(\text{LeakyReLU}(a^\top [Wh_i \,||\, Wh_k]))} \tag{6}$$

This attention mechanism enables the network to focus on the most informative neighbors. The concept of multi-head attention further enhances the model's capacity by simultaneously aggregating information from several independent attention mechanisms:

$$h'_i = \Big\|_{k=1}^{K} \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^k W^k h_j\right) \tag{7}$$

GATv2 [20] an improved version of the original GAT architecture focuses on decoupling the attention computation from the feature transformation step, resulting in greater flexibility and improved performance on diverse and structurally complex graph data.

2.3 Limitations

Graph Neural Networks (GNNs) continue to face several core limitations when applied to complex real-world graphs. Below, we highlight three prominent challenges that hinder their effectiveness:

- **Over-smoothing:**

  Over-smoothing describes the phenomenon where, after repeated message passing, node representations become increasingly similar across the graph. Although early layers improve contextual awareness by integrating information from neighbors, deeper layers tend to blur distinctions between nodes. This homogenization diminishes the model's capacity to preserve discriminative features, ultimately reducing performance in classification and clustering tasks.

- **Under-reaching:**

  Under-reaching reflects the model's inability to capture long-range dependencies due to a limited propagation scope. When the receptive field of a GNN is restricted to immediate neighbors or shallow neighborhoods, the model lacks access to distant yet influential nodes. This can lead to incomplete structural awareness and negatively affect predictions in tasks where non-local interactions are essential.

- **Over-squashing:**

  Over-squashing arises when GNNs attempt to compress information from exponentially large neighborhoods into fixed-size node embeddings. As message paths lengthen in deeper architectures, the network is forced to condense complex and structurally rich information into compact representations. This compression often results in the loss of crucial relational signals, limiting the model's ability to reason about distant nodes and harming performance on tasks requiring high-fidelity structural information.

3 Methodology

3.1 Homophily and Heterophily

The limitations previously described (*over-smoothing*, *under-reaching*, and *over-squashing*) are closely connected to the inherent structural properties of graphs, specifically *homophily* and *heterophily*. Understanding these structural properties is critical for developing effective filtering strategies tailored to different graph types.

Homophily refers to the tendency of nodes to connect predominantly with similar nodes, sharing common attributes or labels. In homophilic graphs, nodes benefit from local aggregation, as neighboring nodes often share similar features, thus simplifying tasks such as node classification. However, excessive neighborhood aggregation in highly homophilic graphs frequently leads to *over-smoothing*, diminishing the distinctiveness of node embeddings and reducing their discriminative power. Spectral filtering for homophilic graphs generally emphasizes low-frequency signals, focusing on smooth patterns within local communities.

Conversely, **heterophily** describes graph structures where nodes tend to connect primarily with dissimilar nodes, resulting in heterogeneous and structurally complex neighborhoods. In such cases, standard message-passing strategies are often insufficient, leading to *under-reaching* and *over-squashing*, as nodes fail to adequately capture essential long-range dependencies and diverse structural nuances. Spectral filtering in heterophilic graphs, therefore, must prioritize higher-frequency signals and diverse spectral bands to capture complex and global relational patterns effectively.

To quantitatively examine the level of homophily or heterophily in graphs, the **homophily indicator (or homophily ratio)** is commonly employed. This indicator measures the proportion of edges connecting nodes of similar labels or features compared to the total edges in the graph. A high homophily ratio indicates strongly homophilic conditions, whereas a lower ratio signals heterophilic structures.

Our research aims to assess and analyze whether graph transformer architectures can effectively mitigate these issues. We specifically investigate the suitability of transformers combined with adaptive band-pass spectral filtering techniques, exploring their potential to address the unique challenges posed by varying degrees of homophily and heterophily in graph data.

3.2 Graph Transformers

Graph Transformers represent a significant evolution in graph representation learning, drawing inspiration from the powerful attention mechanisms originally developed for sequential data in natural language processing. The fundamental similarity between transformers and graphs lies in their reliance on capturing relational dependencies—transformers utilize self-attention to dynamically weigh relationships among sequence elements, analogous to how nodes in graphs interact through their edges. Unlike traditional Graph Neural Networks (GNNs), which often suffer from limitations such as over-smoothing, under-reaching, and over-squashing due to rigid message-passing schemes, Graph Transformers dynamically adjust their attention to relevant nodes and edges across diverse graph structures. This adaptability enables transformers to effectively handle varying structural complexities inherent to graph data.

3.2.1 Core Architecture

The fundamental architecture of Graph Transformers comprises several layers, as illustrated in figure **??**of self-attention mechanisms, positional encodings, feed-forward and normalization :

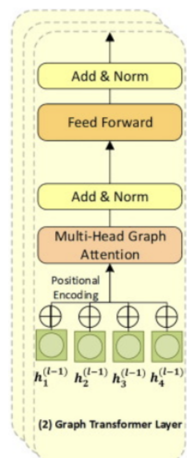


Figure 1: Overview of graph transformer architecture.

- **Self-Attention Mechanism:** capable of effectively capturing long-range dependencies without relying on recurrent or convolutional structures. Our base transformer architecture incorporates graph structural biases directly into the attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + B\right) V \tag{8}$$

Here, Q, K, and V represent queries, keys, and values from node embeddings, while B encodes graph connectivity information.

Multi-head attention enhances the model by simultaneously attending to different node relationships through parallel attention computations, helping capture both local and global graph structures effectively.

- **Positional Encodings:** Inject positional or structural information into node embeddings to retain awareness of the underlying graph topology, allowing transformers to distinguish structurally similar nodes.

- **Normalization (Add & Norm):** Includes residual connections followed by layer normalization to stabilize training, mitigate vanishing or exploding gradients, and maintain stable gradient flows throughout deep transformer layers.

3.2.2 Graph Transformer Varieties

Several widely recognized Graph Transformer architectures include:

- **Graphormer [21]:** Designed explicitly for graph classification tasks, Graphormer employs sophisticated structural encodings (spatial and edge encodings) to effectively capture global graph context, notably excelling in molecular classification.

- **Spectral Graph Transformers(e.g., GraphSAN) [27]:** These transformers incorporate spectral filtering to exploit frequency-domain information of graph structures, demonstrating strong capabilities in tasks sensitive to global structural patterns, such as anomaly detection and link prediction.

Given the central role of structural awareness in graph representation, selecting an appropriate method for positional encoding becomes a key design choice in transformer-based graph models.

3.2.3 Methods for Positional Encoding

Positional encoding methods play a crucial role in Graph Transformers performance since they are responsible of embedding structural information into node representations:

- **Laplace Eigenvector Positional Encoding (LEPE):** Our research employs Laplacian eigenvectors derived from graph Laplacians as positional encodings. LEPE captures intrinsic global structural properties of the graph, aiding in differentiating nodes based on their positions in the overall graph topology.

- **Random Walk Positional Encoding (RWPE):** An alternative method commonly used in graph transformers, RWPE encodes positional information based on random walk probabilities. This method effectively captures local and global structure by measuring node connectivity and accessibility within random walk contexts.

4 Experiments and Results

4.1 Empirical Observations

At the beginning of this project, preliminary experiments were performed to assess the effectiveness of various state-of-the-art graph neural network models on two benchmark datasets: one predominantly homophilic (*Cora*, homophily ratio = 0.81) and one strongly heterophilic (*Texas*, homophily ratio = 0.11).

The *Cora* dataset [28] is a well-known citation network commonly used for node classification tasks, comprising scientific publications represented as nodes, and citations among them represented as edges. Nodes are categorized into seven distinct research topics, exhibiting strong homophily since connected papers often belong to similar research categories.

Conversely, the *Texas* dataset [30] is derived from university webpage networks, representing individual webpages as nodes and hyperlinks as edges. It is characterized by strong heterophily, as linked webpages often differ substantially in content and category, leading to significantly diverse connections.

Table 1: Performance of Various Graph Models on Cora and Texas Datasets

Model	Cora (Accuracy %)	Texas (Accuracy %)
GCN	80.4	50.1
GIN	71.2	62.2
GAT	76.2	72.9
GraphSan	76.4	72.2
Graphormer	76.9	72.9

As shown in Table 1, transformer-based models exhibit promising performance on the strongly heterophilic Texas dataset. These observations motivate to investigate further into whether transformer architectures possess inherent capabilities such as band-pass filtering or other mechanisms that enable them to effectively handle heterophilic conditions across diverse scenarios.

Interestingly, recent findings by Sáez de Ocáriz Borde et al. [31] suggest that transformer models may effectively handle heterophilic graphs without necessarily relying on attention mechanisms, emphasizing instead the critical role of normalization components. Motivated by these insights and aiming to understand the relative contributions of different graph transformer components, we conducted a comprehensive ablation study.

4.2 Ablation Study

To rigorously evaluate the importance of individual transformer components, we adopted an ablation study methodology. All models were trained under consistent conditions: identical datasets (Cora and Texas), an equal number of training epochs, and similar hyperparameter settings to ensure fairness and stability in comparisons.

For performance evaluation, we primarily considered two key metrics: the test accuracy of each model and the frequency response of the learned filters, reflecting the model's spectral filtering capabilities.

4.3 Ablation Study Design

Our base model includes all components; positional encoding, transformer layers, normalization, and linear layers, and is initially tested on both the Cora and Texas datasets. We then evaluate the performance impact of various combinations of these components. When specific components are omitted, the attention mechanism is replaced with a GCN layer to serve as a baseline. Specifically, when the attention layer is excluded, we replace it with a standard GCN convolutional layer.

The detailed structure of our ablation experiments is summarized in Table 2:

Table 2: Architecture Ablation Study and Results

Postional Encoding	TransformersLayer	Norm	Cora Acc. (%)	Texas Acc. (%)
✓	✓	✓	52.0	62.1
✓	✗	✗	77.5	59.0
✗	✓	✓	53.8	60.5
✗	✓	✗	54.1	58.7
✗	✗	✓	55.4	57.3
✓	✓	✗	58.3	60.8
✗	✗	✗	80.4	47.1

4.4 Interpretation of Ablation Results

The ablation study in Table 2 evaluates the contributions of positional encoding (PE), Transformer layers, and normalization (Norm) to graph neural network performance across two structurally distinct datasets: Cora (homophilic) and Texas (heterophilic). The results offer clear insight into the architectural needs of different graph types.

On the Cora dataset, which exhibits strong homophily, the vanilla GCN achieves the highest accuracy (80.4%), reaffirming its effectiveness in low-frequency-dominated graphs [32]. The strong performance aligns with GCN's known low-pass filtering behavior, which smooths features across neighboring nodes that share similar labels, as illustrated in Figure 2a.

However, this performance degrades substantially on Texas (47.1%), a heterophilic dataset where local neighborhood aggregation leads to over-smoothing and loss of class-discriminative signals [33], as shown in Figure 2b.

In contrast, the full Transformer-based model with positional encoding and normalization achieves the best result on Texas (62.1%), despite yielding poor performance on Cora (52.0%). This indicates that global attention mechanisms and structure-aware encodings are more effective in heterophilic settings. The Laplacian-based positional encoding contributes structural information beyond local neighborhoods [34, 35], allowing the model to distinguish nodes based on broader positional context. Normalization (LayerNorm) appears to regularize node feature distributions, enhancing model stability [36].

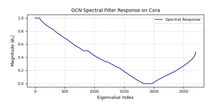
Interestingly, the configuration using only positional encoding performs second-best on Cora (77.5%) and moderately on Texas (59.0%), highlighting that even without attention, graph-aware features contribute significantly to performance. This also suggests that lightweight components like PE can serve as powerful structural signals when the graph topology is homophilic.

In summary, these findings support the hypothesis that no single architecture is universally optimal. Instead, model design should align with the graph's structural properties. Global architectures combining Transformer attention, structural encoding, and normalization show the most robust generalization on non-homophilic graphs.

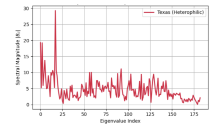
4.5 Spectral Filtering Behavior and Interpretation

To further understand the performance dynamics observed in the ablation study, we examine the spectral filter responses of each architecture by projecting their learned embeddings onto the Laplacian eigenbasis. These projections highlight how each model modulates different graph frequency components.

In homophilic graphs such as Cora, the GCN exhibits strong low-pass filtering, with its spectral response peaking at low eigenvalue indices and decaying across higher frequencies (Figure 2a). This behavior aligns with the observed high accuracy of GCN on Cora, as smoothing enhances label consistency in local neighborhoods. On the heterophilic Texas dataset, however, GCN's spectral response becomes irregular and noisy (Figure 2b), reflecting its inability



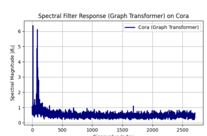
(a) GCN frequency response on Cora



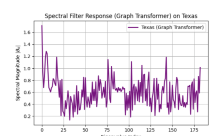
(b) GCN frequency response on Texas

Figure 2: Spectral filter responses of GCN on (a) Cora (homophilic) and (b) Texas (heterophilic).

to selectively amplify or suppress frequencies relevant to class separation. In contrast, the Graph Transformer demonstrates a broader and more distributed spectral response on Texas (Figure 3b), suggesting band-pass filtering capabilities that allow the model to extract both global and local signals. This aligns with its superior performance on Texas, where class information is often embedded in higher frequencies.



(a) Transformer frequency response on Cora



(b) Transformer frequency response on Texas

Figure 3: Spectral filter responses of Graph transformer on (a) Cora (homophilic) and (b) Texas (heterophilic).

Interestingly, on Cora, the Graph Transformer produces a distinct spectral profile with sharp peaks at low frequencies followed by a flattened distribution (Figure 3a). While this retains low-frequency sensitivity, it also suggests that Transformers are capable of avoiding oversmoothing by preserving mid-frequency content—though this comes at the cost of optimality in homophilic settings.

Overall, the spectral responses reinforce the architectural findings: GCNs are well-suited for low-frequency tasks (homophily), while Graph Transformers provide flexible, band-pass-like filtering behavior that enhances generalization on heterophilic graphs.

4.6 Theoretical Justification

The performance differences observed across architectures can be explained by their interaction with the structural properties of the graph. GCNs are inherently suited for homophilic graphs due to their neighborhood aggregation scheme, which operates as a *low-pass filter*:

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{9}$$

This formulation smooths features across adjacent nodes, which improves performance when neighboring nodes share the same label, as in Cora. However, in heterophilic graphs like Texas, this leads to *oversmoothing*, where distinct node features become indistinguishable.

Transformer layers, by contrast, do not rely on local neighborhood aggregation and are capable of capturing *long-range dependencies*. When combined with graph-based positional encodings (e.g., Laplacian eigenvectors), they allow the model to incorporate global structural information beyond immediate connections.

Normalization layers (such as LayerNorm) additionally help stabilize feature distributions during training and have been shown to enhance performance, particularly in structurally diverse and heterophilic settings.

Conclusion

Conclusion and Future Directions

In this work, we explored how specific graph transformerarchitectural elements such as positional encoding, attention layers, and normalization could affect the performance of graph neural networks on graphs with differing structural properties. Through a series of ablation experiments, we observed that traditional GCNs maintain strong performance in homophilic settings such as Cora, but their effectiveness diminishes on heterophilic graphs like Texas. In contrast, Transformer-based models, particularly those incorporating structural encodings and normalization, demonstrated better generalization on heterophilic data, suggesting their adaptability to more complex connectivity patterns.

To support and further interpret these results, we analyzed the spectral characteristics of each model by examining their responses in the frequency domain. This revealed that GCNs primarily emphasize low-frequency components, aligning with their neighborhood-averaging behavior. However, such low-pass filtering tends to obscure meaningful variations in heterophilic graphs. On the other hand, Graph Transformers exhibited broader and more balanced spectral responses, indicative of band-pass filtering behavior capable of capturing both local and global structural signals. These spectral findings prove the empirical performance differences and highlight the role of frequency-sensitive architectures in heterogeneous graph settings.

While these insights are promising, several aspects merit further research. First, the spectral observations point to the need for a more formal understanding of how frequency-domain behavior correlates with model performance across various graph types. A theoretical framework describing these relationships could inform the design of more adaptive GNN architectures. Second, expanding this study to include a wider range of datasets covering both homophilic and heterophilic extremes would provide a more robust evaluation of generalization. Additionally, given that some of the Transformer variants explored here are relatively compact, future work could assess their efficiency in resource-constrained conditions, contributing to the development of lightweight graph learning systems aligned with Green AI objectives. Finally, validating these findings on large-scale and real-world graphs remains a critical next step to assess the scalability and practical utility of these approaches.

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